Muhammad Adnan Haseeb

Full Stack AI Engineer | DevOps

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Professional Summary

Full stack AI engineer building and running AI-assisted finance and data workflows in production: LLM document extraction checked by deterministic validation, invoice and statement reconciliation, human review and approval queues, and integrations with Microsoft 365 (Graph API, Entra ID), Zoho Books and WhatsApp. Works across Python and Kotlin microservices on GCP, and deploys and runs services on AWS with CI/CD, feature flags, kill switches and tests on every change. Pairs every AI output with code-level validation, and routes anything uncertain to human or rule-based approval before it reaches the ledger.

Experience

StechAI | Lahore, Pakistan **Jul 2025 - Present**
Full Stack AI Engineer
Backend systems, REST APIs, integrations, AI workflows and deployments for client products, from build and testing through production support.

BRKZ (Saudi B2B marketplace), supplier finance automation
Python, Kotlin, FastAPI, Spring Boot, MongoDB, LLMs, GCP, Zoho Books API, Meta WhatsApp Cloud API

- Made **LLM-based invoice extraction and purchase-order matching** more reliable by adding deterministic validation around model output, catching e-invoice identifiers misread as invoice numbers on about 12% of one major supplier's invoices before they reached the ERP.
- Developed **statement-of-account reconciliation**: supplier statements are read with LLMs, every row is matched against the ledger by invoice number, then by amount and date, and suppliers receive a generated PDF of the exact invoices missing.
- Built an **AI agent that automates supplier document collection** for VAT reporting: scheduled WhatsApp reminders with quick-reply buttons that pause the chase, working-day-aware scheduling, and escalation to a voice agent with full conversation context.
- Delivered production integrations across Python and Kotlin microservices (GCP Pub/Sub, Zoho Books API with webhooks, WhatsApp Cloud API) with feature flags, kill switches, idempotent messaging and tests on every change.

ClinFocus, clinical data reconciliation
AWS EC2, systemd, nginx, GitHub Actions, Microsoft 365 Graph API, Python, TypeScript, FastAPI, MongoDB, LLMs

- Deployed and maintained the platform on **AWS EC2** with systemd, nginx and GitHub Actions CI/CD, and cut a single rule run from 4.1 minutes to under a minute by caching extraction on content rather than on generated file bytes.
- Integrated **Microsoft 365 through the Graph API**: Entra ID app registration and the OAuth2 client-credentials flow to send transactional mail from a designated mailbox, with SMTP and log transports chosen by configuration and a clear error when SMTP AUTH is disabled on the tenant.
- Built a five-stage **data reconciliation pipeline** that matches 276 collected fields to their expected schema, extracts 14 normalised datasets from 69 source forms, reconciles the values, and works out which of 86 validation rules a study has the columns to run before any execute.
- Delivered the **review workspace** where every flagged record becomes a tracked finding with status, approval, site response and history, plus bulk actions, per-rule filters, CSV export and a model log that separates rules that ran clean from rules that never read the column they needed.

Freelance Software Developer | Fiverr | Remote **Mar 2023 - Present**

- Build web applications, backend services and REST APIs to client requirements, integrating databases and third-party APIs, and maintain and debug existing applications.

Technical Skills

Finance and data workflows: invoice extraction, invoice-to-PO matching, statement-of-account reconciliation, VAT document tracking, review and approval queues, CSV export

AI / LLM: LLM document extraction, structured output with schema validation, deterministic checks on model output, AI agents, prompt engineering, OpenAI SDK, Pydantic

Integrations: Microsoft 365 (Graph API, Entra ID app registration, OAuth2), Zoho Books API, Meta WhatsApp Cloud API, REST APIs, webhooks

Backend and frontend: Python, FastAPI, Django, Kotlin, Spring Boot, Node.js, Express, TypeScript, React, Next.js

Data and ML: MongoDB, PostgreSQL, MySQL, Pandas, PyTorch, scikit-learn

Cloud and DevOps: GCP (GKE, Pub/Sub, Cloud Storage, Secret Manager), AWS EC2, Docker, nginx, systemd, GitHub Actions, Datadog

Reliability and testing: pytest, feature flags, kill switches, idempotent processing, safe rollout

Education

Bachelor of Computer Science, Arid Agriculture University Rawalpindi (Burewala Campus), CGPA 3.46 / 4.0 **2021 - 2025**